Exp No: 3.a Conduct an experiment to show data visualization using line plot

AIM:

To conduct an experiment to show data visualization using line plots.

PROGRAM:

```
import matplotlib.pyplot as plt

Months = list(range(1, 11))  # Months 1 to 10

Company_A = [130, 135, 152, 138, 160, 165, 172, 180, 178, 200]

Company_B = [100, 108, 105, 115, 130, 138, 145, 140, 152, 170]

plt.plot(Months, Company_A, color="purple", label="Company A (Tech)")

plt.plot(Months, Company_B, color="blue", label="Company B (Auto)")

plt.title("Stock Price Trend Comparison")

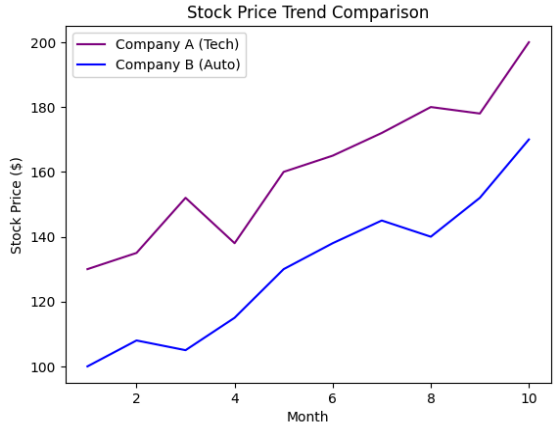
plt.xlabel("Month")

plt.ylabel("Stock Price ($)")

plt.legend(loc="upper left")

plt.show()
```

OUTPUT:



RESULT:

Hence an experiment to show data visualization using line plot is done.

Exp No: 3.b Conduct an experiment to show data visualization using bar chart

AIM:

To conduct an experiment to show data visualization using a bar chart.

PROGRAM:

```
import matplotlib.pyplot as plt

import numpy as np

Categories = ['ELECTRONICS', 'CLOTHING', 'HOME DECOR', 'BOOKS']

xaxis = np.arange(len(Categories))

Sales = [12500, 8400, 11000, 3100]

bars = plt.bar(Categories, Sales)

plt.xticks(xaxis, Categories, rotation=45)

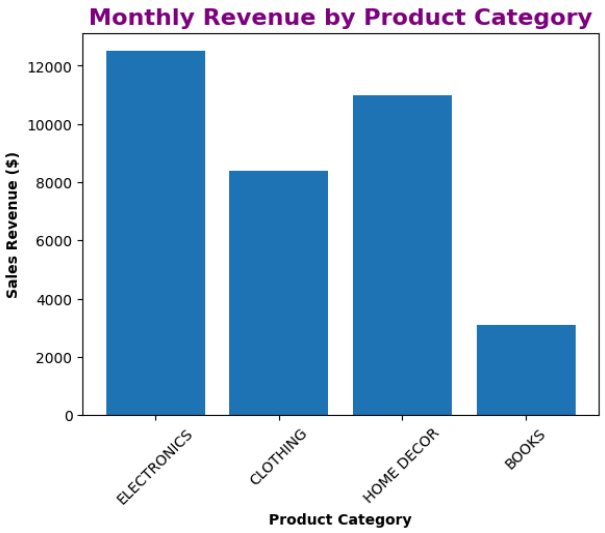
plt.xlabel("Product Category", fontweight="bold")

plt.ylabel("Sales Revenue ($)", fontweight="bold")

plt.title("Monthly Revenue by Product Category", fontsize=16, color="purple", fontweight="bold")

plt.show()
```

OUTPUT:



RESULT:

Hence an experiment to show data visualization using a bar chart is done.

Exp No: 3.c Conduct an experiment to show data visualization using pie chart

AIM:

To conduct an experiment to show data visualization using pie charts.

PROGRAM:

```
import matplotlib.pyplot as plt

labels = ['Android', 'iOS', 'Windows', 'Others']

shares = [70.5, 32.2, 15.5, 9.5]

colors = ['purple', 'skyblue', 'pink', 'darkblue']

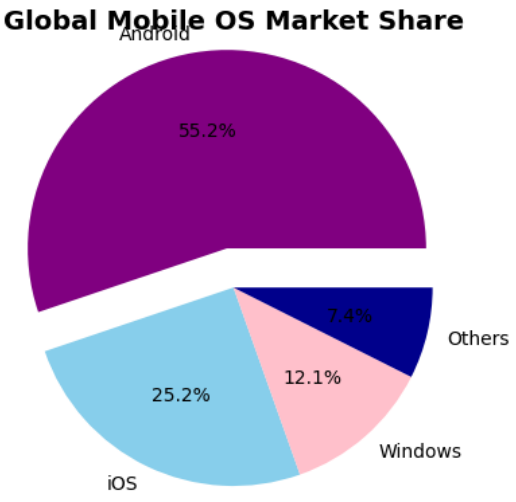
explode = (0.2, 0, 0, 0)  # Explode the leading slice (Android)

plt.pie(shares, labels=labels, colors=colors, explode=explode,
autopct='%0.1f%%')

plt.title('Global Mobile OS Market Share', fontsize=14, fontweight='bold')

plt.show()
```

OUTPUT:



RESULT:

Hence an experiment to show data visualization using pie charts is done.